
ST-SIMDIFF: BALANCING SPATIOTEMPORAL SIMILARITY AND DIFFERENCE FOR EFFICIENT VIDEO UNDERSTANDING WITH MLLMs

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ABSTRACT

Multimodal Large Language Models (MLLMs), when confronted with long-form video, incur prohibitive computational overhead that stems directly from the sheer volume of visual tokens their processing pipeline demands. In pursuit of greater efficiency, prevailing methods have predominantly sought to attenuate redundancy by pruning or merging tokens according to criteria of importance or similarity. Such approaches, however, largely disregard a dimension of video content that is arguably indispensable, namely the changes and turning points that structure an unfolding narrative, and they furthermore lack any collaborative formulation of spatio-temporal relationships. To redress these shortcomings, we advance a reframed perspective in which similarity serves to expose redundancy whereas difference serves to apprehend salient events. Grounded in this dichotomy, we devise a training-free framework, termed ST-SimDiff. We begin by constructing a spatio-temporal graph over the visual tokens so as to model their intricate associations within a single unified representation. Thereupon we deploy a parallel dual-selection strategy comprising two complementary mechanisms: (1) similarity-based selection, which leverages community detection to retain representative tokens and thereby compress static information; and (2) temporal difference-based selection, which precisely localizes content-changing points in order to preserve those tokens that encode pivotal dynamic shifts. This design enables the framework to conserve both static and dynamic content while committing only a minimal token budget. Extensive experimentation demonstrates that our method substantially outperforms state-of-the-art approaches even as it markedly curtails computational cost. Our code will be released upon acceptance.

1 INTRODUCTION

The ascendancy of Large Language Models (LLMs) has substantially propelled the development of Large Vision-Language Models (LVLMs), which have come to exhibit remarkable proficiency across both image and video understanding (Zhang et al., 2024c; Liu et al., 2024c). For the purpose of video processing, contemporary LVLMs conventionally sample a video into a sequence of discrete frames and subsequently transcribe each frame into hundreds or even thousands of visual tokens destined for the LLM (Bai et al., 2025; Li et al., 2024). Efficacious though this paradigm may be, the token count escalates explosively as video duration and resolution increase, precipitating computational and storage burdens so severe as to be prohibitive, and thereby sharply circumscribing the deployment of LVLMs in settings such as long-video analysis and real-time interaction (Fu et al., 2024b).

Confronted with this challenge, a diverse array of methods has been proposed with a view to enhancing the efficiency of LVLMs, and these may be broadly partitioned into two categories. The first is importance-based pruning. As FastV (Chen et al., 2024) observes, the attention distribution across the model’s successive layers is marked by pronounced redundancy when LVLMs process visual information; visual tokens that accrue lower attention scores in the deeper layers may consequently be pruned in order to alleviate the computational load. The second category comprises similarity-based merging and selection. Prior work has established that visual tokens display considerable mutual similarity, whether between temporally adjacent frames or within the confines of

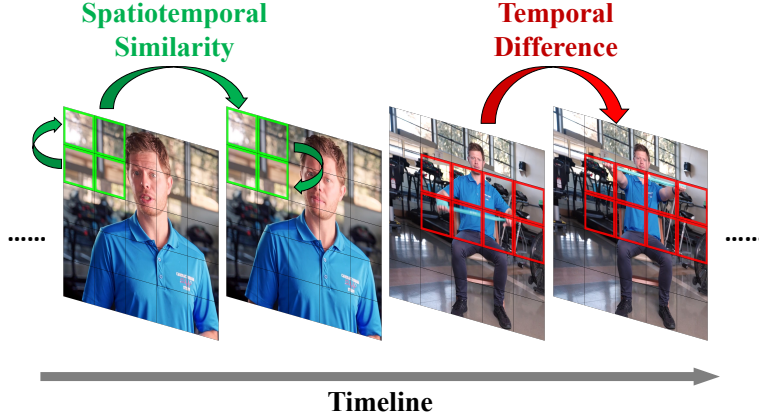


Figure 1: The core motivation underlying ST-SimDiff. We contend that efficient video understanding necessitates the simultaneous treatment of two complementary regimes: Spatiotemporal Similarity (left), which may be exploited to compress redundant information both spatially within a frame and temporally across adjacent frames; and Temporal Difference (right), which ought to be detected so as to apprehend the salient actions or events that constitute the plot.

a single frame. Accordingly, FrameFusion (Fu et al., 2024b) attenuates redundancy by merging similar tokens drawn from adjacent frames, whereas VisionZip (Yang et al., 2025) instead selects dominant tokens directly at the level of the vision encoder.

Notwithstanding the success that existing methods have attained in visual token reduction, they remain encumbered by two salient limitations. First, prevailing token pruning methods tend to concentrate either upon spatial correlations confined to a single frame or upon temporal correlations at identical positions across distinct frames; this absence of any joint modeling and analysis of spatio-temporal correlations precludes them from effectively capturing complex dynamic events. Beyond this, such methods share a latent blind spot, in that they attend chiefly to informational commonalities—similarity and importance foremost among them—while neglecting the changes and differences that prove decisive. Because the narrative of a video is so frequently propelled by transitional events, a model that pursues similarity alone is liable to smooth over an abrupt action and, in so doing, to misconstrue the underlying content.

Motivated by these considerations, we advance a reframed perspective in which similarity is enlisted for the identification of redundancy while difference is enlisted for the apprehension of pivotal moments. We hold that an ideal video token compression algorithm ought to accomplish two objectives concurrently: to render the stable content of a video with the fewest possible tokens, and to preserve, with precision, the key changes it exhibits. To this end, we propose **ST-SimDiff**, a dual token selection framework predicated upon a spatio-temporal graph. Concretely, the framework is composed of two constituents. The first is **similarity-based representative token selection**, in which we regard all visual tokens of a video as nodes and construct a graph grounded in their feature similarity along the spatio-temporal dimension; by means of a graph community detection algorithm, we isolate tightly connected token clusters that stand for the stable and persistent visual elements of the video, and from each cluster we retain a small number of tokens distinguished by higher centrality to serve as representatives. The second is **difference-based event token selection**, in which we direct particular scrutiny toward the edges that run along the temporal axis of the graph: whenever the similarity between corresponding tokens of adjacent frames declines abruptly, we construe this as a turning point and designate the implicated tokens as critical event tokens that must be retained. Finally, we amalgamate these two token sets before feeding them into the LLM. Our contributions can be summarized as follows:

- We are the first to contend that the similarity and the difference of tokens warrant equal consideration within research on VLM efficiency.
- We devise a spatio-temporal graph framework that models, within a single unified representation, the complex spatio-temporal relationships that obtain among video tokens, and

we accordingly propose a novel dual token selection strategy capable of screening simultaneously for both representative and pivotal event tokens.

- We undertake extensive experimentation across video understanding benchmarks, the results of which establish that ST-SimDiff markedly curtails the visual token budget while sustaining, or indeed improving upon, task performance.

2 RELATED WORK

2.1 LARGE VISION LANGUAGE MODELS

The rapid advancement of Large Language Models (LLMs) (Touvron et al., 2023; Achiam et al., 2023; Ouyang et al., 2022) has appreciably accelerated progress in multimodal understanding, thereby occasioning the emergence of numerous prominent Large Vision Language Models (LVLMs) (Liu et al., 2024b; Bai et al., 2023). These LVLMs characteristically process images or video frames by transducing them into visual tokens through pre-trained visual encoders, whereupon the resulting tokens are relayed to the LLMs. The visual encoder and the LLM are commonly bridged by a variety of alignment modules, among them MLPs (Liu et al., 2024a;b) and Q-Formers (Li et al., 2023; Dai et al., 2023). Such an architecture endows LVLMs with exceptional competence across multimodal tasks (Xi et al., 2025; Holdaway et al., 2024; Zhang et al., 2025a), and in the domain of video understanding in particular they evince considerable promise in the processing of longer, higher-resolution, and correspondingly more complex videos (Bai et al., 2025; Li et al., 2024; Zhang et al., 2024c; Liu et al., 2024c).

This formidable capability, however, is not without significant computational cost, a cost that manifests most acutely in video understanding scenarios. Owing to the imperative of processing continuous sequences of frames, the number of visual tokens proliferates exponentially and far outstrips that engendered by static images, thereby posing a severe impediment to both the training and the inference efficiency of LVLMs. Consequently, the question of how to process massive video visual tokens efficiently while curtailing computational overhead persists as a critical concern in contemporary LVLM research.

2.2 VISUAL TOKEN COMPRESSION

Impelled by the substantial overhead intrinsic to video processing, visual token compression has emerged as a critical avenue for improving the efficiency of Large Visual Language Models (LVLMs). Extant methods approach the problem of compression predominantly from two complementary vantage points, namely token importance and token similarity.

Earlier methods concentrated principally upon importance-based token pruning, the aim of which is to identify and excise those visual tokens that contribute least to model performance. Such methods typically enlist attention scores or other feature-derived metrics in order to quantify token importance, as exemplified by FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b). Subsequent works (Shang et al., 2024; Ye et al., 2025) have begun to incorporate token similarity as well, the better to attenuate redundancy while preserving critical information; these methods acknowledge that even tokens of high importance may nonetheless exhibit pronounced similarity, and hence engender informational redundancy. FrameFusion (Fu et al., 2024b), for instance, conjoins similarity-based merging with importance-based pruning; VisionZip (Yang et al., 2025) diminishes redundancy by selecting dominant tokens and thereupon merging contextual information; and VisPruner (Zhang et al., 2025b) harnesses visual cues to expunge redundant items.

Despite the considerable progress achieved in the treatment of visual token redundancy, existing work scarcely engages with the intricate, global spatio-temporal similarity relationships that obtain among tokens. In video scenarios, visual information is redundant along both the spatial and the temporal axis. Existing similarity-driven methods, FrameFusion among them, attend predominantly to the temporal similarity between adjacent frames alone, and thereby fail to exploit the complex, jointly spatio-temporal information available to them. This shortfall constrains their capacity to fully capitalize on the redundancy inherent in video and, in turn, curtails their performance under regimes of extreme token reduction.

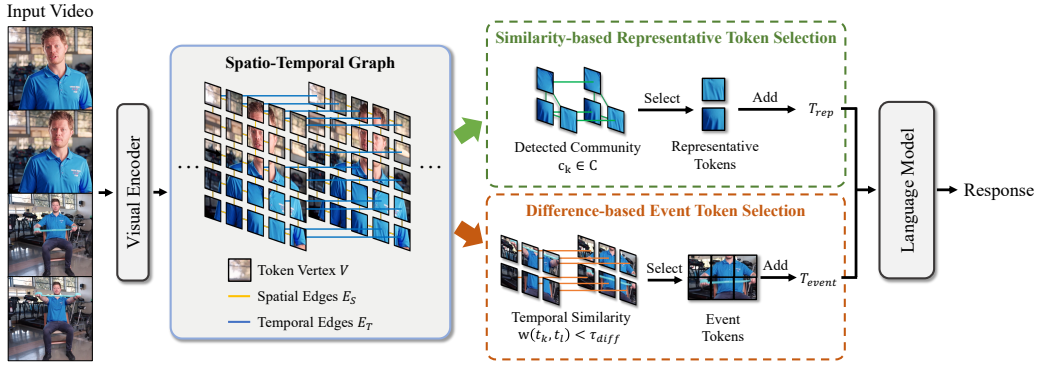


Figure 2: An overview of the ST-SimDiff framework, which is articulated into three constituent parts: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection, and Difference-based Event Token Selection.

3 PROBLEM DEFINITION

Given a video V together with a text query Q , a standard Large Vision-Language Model (LVLM) first encodes the video into a complete sequence of N visual tokens, $T_{\text{full}} = \{t_1, t_2, \dots, t_N\}$. This encoding is computationally demanding, inasmuch as the complexity of the self-attention mechanism scales quadratically with the token count N , that is, as $O(N^2)$. Our task is therefore to devise an efficient token selection function, $f(\cdot)$, that receives the full sequence T_{full} alongside a compression ratio r and yields a substantially smaller subset $T_{\text{sub}} = f(T_{\text{full}}, r)$. The objective is to maximize the LVLM’s downstream task performance upon this subset, subject to the constraint that its cardinality $|T_{\text{sub}}|$ approximate $r \cdot N$. The crux of the matter, accordingly, is to design the selection function $f(\cdot)$ in such a manner that the most critical information is retained within the subset T_{sub} .

4 ST-SIMDIFF

4.1 METHOD OVERVIEW

As illustrated in Figure 2, the proposed ST-SimDiff method is constituted by two key components that operate in parallel upon a spatio-temporal graph constructed from the video’s visual tokens: Similarity-based Representative Token Selection (SRTS) and Difference-based Event Token Selection (DETS). First, within the SRTS module, we identify and condense the stable and redundant content of the video. By applying community detection algorithms to the graph, we aggregate highly similar tokens into clusters that correspond to persistent visual elements such as static backgrounds, and from each cluster we thereupon select a small number of highly central tokens so as to constitute a representative set, T_{rep} . Second, the DETS module is contrived to apprehend the crucial dynamic shifts of the video. It scrutinizes the temporal edges of the graph and thereby localizes those moments at which the similarity between corresponding tokens in adjacent frames declines abruptly; such points of pronounced difference betoken turning points, and the tokens that frame these transitions are preserved as critical event tokens, forming the set T_{trans} . By combining these two components synergistically, ST-SimDiff produces a final token subset $T_{\text{sub}} = T_{\text{rep}} \cup T_{\text{trans}}$ that is at once highly compact and richly informative, retaining stable content even as it captures key events with precision.

4.2 SPATIO-TEMPORAL GRAPH CONSTRUCTION

Given a video, the vision encoder transforms it into a sequence of tokens $T = \{t_1, t_2, \dots, t_N\}$. Each token $t_k \in T$ corresponds to a particular spatio-temporal location and is associated with a feature vector $\mathbf{x}_k$. We characterize the location of each token t_k by way of three indices: its frame index $T(t_k)$, its spatial height index $H(t_k)$, and its spatial width index $W(t_k)$.

We model the relationships that subsist among these tokens by means of a spatio-temporal graph $G = (V, E)$, in which the vertex set V stands for all tokens in T . The edge set E is formed as the union of two distinct subsets, $E = E_S \cup E_T$, which capture spatial and temporal relationships respectively. The spatial edges E_S connect tokens that are spatially adjacent within one and the same frame:

$$E_S = \{(v_i, v_j) \in V \times V \mid T(t_i) = T(t_j) \text{ and } |H(t_i) - H(t_j)| + |W(t_i) - W(t_j)| = 1\} \quad (1)$$

The temporal edges E_T , for their part, connect tokens that occupy the same spatial position but reside in adjacent frames, and thereby capture temporal continuity:

$$E_T = \{(v_i, v_j) \in V \times V \mid H(t_i) = H(t_j), W(t_i) = W(t_j), \text{ and } |T(t_i) - T(t_j)| = 1\} \quad (2)$$

The weight of any edge $w(v_i, v_j) \in E$ is defined as the cosine similarity between the feature vectors of the corresponding tokens:

$$w(v_i, v_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \quad (3)$$

This sparse graph structure encodes local spatial relationships and temporal continuity with considerable efficiency, and in so doing furnishes a unified foundation upon which our dual-path token selection strategy is erected.

4.3 SIMILARITY-BASED REPRESENTATIVE TOKEN SELECTION

Video content is intrinsically characterized by substantial spatio-temporal redundancy. In a static scene that persists for several seconds, for instance, the background tokens exhibit pronounced similarity over time; likewise, for an object traversing the screen, its corresponding tokens, notwithstanding their shifting spatial positions, sustain a high degree of semantic correlation. Whereas existing methods frequently treat temporal and spatial similarities in isolation from one another, our spatio-temporal graph is expressly designed to capture these similarities jointly. Within this graph, semantically related tokens coalesce into dense “communities” or “clusters” on the basis of their feature similarity, and do so irrespective of the particular frame or location in which they arise. It follows that the selection of representatives from each community constitutes a highly efficient strategy for compressing redundant information while preserving the core semantics intact.

In order to identify strongly correlated token clusters with precision, we first subject the graph G to a thresholding operation. We prescribe a similarity threshold τ_{sim} and retain solely those edges whose weights surpass this threshold, thereby forming a new graph $G' = (V, E')$, where $E' = \{(v_i, v_j) \in E \mid w(v_i, v_j) > \tau_{\text{sim}}\}$. We then apply a graph community detection algorithm—the Louvain method (Blondel et al., 2008), for example—to G' so as to identify token clusters $C = \{c_1, c_2, \dots, c_m\}$. For each community $c_k \in C$, we proceed to rank and filter its constituent tokens according to their centrality. The centrality score $S_c(t_a)$ of a token $t_a \in c_k$ is defined as its mean similarity with respect to all other tokens residing within the community:

$$S_c(t_a) = \frac{1}{|c_k| - 1} \sum_{t_b \in c_k, b \neq a} w(t_a, t_b) \quad (4)$$

Subsequently, we equate the intra-community retention rate with the globally prescribed compression ratio r , and from each community c_k we preserve the top $\lceil |c_k| \cdot r \rceil$ tokens ranked by centrality score. This uniform filtering procedure accommodates all communities without special provision, single-node communities composed of unique tokens included, in which latter case the sole node is retained. Formally, the set of representative tokens, T_{rep} , is defined as:

$$T_{\text{rep}} = \bigcup_{k=1}^m \text{TopK}_{t \in c_k, \text{score} = S_c(t)}(|c_k| \cdot r) \quad (5)$$

where the TopK function returns, from a given set, the subset of k elements bearing the highest scores. In sum, this strategy compresses the static and persistent content of the video efficiently, doing so by identifying and retaining the most central tokens within each semantic cluster.

4.4 DIFFERENCE-BASED EVENT TOKEN SELECTION

If similarity delineates the “norm” of a video, then difference delineates its “events”. A model attuned to similarity alone may excel at discerning “what is” yet founder upon “what happened”. The plot of a video is propelled by key events, and the essence of any event is change—the appearance of a new object, the inception of an action, or a transition between scenes. Within our model, these changes are made manifest as a sharp difference in the features of temporally adjacent tokens. It is therefore of paramount importance to capture, with accuracy, these “turning points” of significant difference, for they are indispensable to understanding the dynamic content of the video and to answering correctly such questions as “when” and “why”.

Difference, and most conspicuously difference along the temporal dimension, frequently heralds the occurrence of a key event. The entry of a new object, a scene change, or the commencement or termination of an action, for instance, will each induce drastic alterations in the visual features at corresponding positions across adjacent frames. We therefore direct our analysis specifically toward the temporal edges (E_T) of our spatio-temporal graph. For any temporal edge $(v_k, v_l) \in E_T$ in the graph, which connects two temporally adjacent tokens t_k and t_l , we prescribe a dynamic threshold τ —for instance, the 95th percentile of all temporal edge difference scores. Whenever the difference score of a temporal edge (v_k, v_l) exceeds this threshold, that is, whenever $w(t_k, t_l) < \tau_{\text{diff}}$, we regard the subsequent token t_l as a critical event token and retain it accordingly. Formally, the set of event tokens, T_{event} , is defined as:

$$T_{\text{event}} = \{t_l \mid \exists t_k \text{ s.t. } (v_k, v_l) \in E_T, T(t_l) > T(t_k), \text{ and } w(v_k, v_l) < \tau_{\text{diff}}\} \quad (6)$$

By virtue of this strategy, those tokens that signify moments of significant temporal change are duly preserved. In capturing these key transitions, we ensure that the crucial dynamic facets of the video’s narrative are not inadvertently overlooked in the course of the compression process.

4.5 OVERALL REDUCTION PROCESS

Within the proposed framework, we first compute the representative token set T_{rep} and the event token set T_{event} in parallel and take their union, $T_{\text{candidate}} = T_{\text{rep}} \cup T_{\text{event}}$, to serve as the initial candidate set. In order to meet the target token count $N_{\text{target}} = \lceil r \cdot N \rceil$ precisely, we introduce a concluding pruning step: should the cardinality of the candidate set, $|T_{\text{candidate}}|$, exceed N_{target} , we excise the $|T_{\text{candidate}}| - N_{\text{target}}$ tokens of lowest importance. In keeping with existing work (Chen et al., 2024), the importance of a token is adjudged by its attention score in the shallow layers of the LLM. This step guarantees that our method can accommodate any computational budget with flexibility while nonetheless preserving critical information.

In sum, our overall method adroitly conjoins two strategies, namely graph-based token selection and attention-based dynamic pruning. The former exploits the intrinsic structure of the video content—its similarity and its difference alike—so as to ensure the retention of core information, whereas the latter affords a flexible mechanism by which any prescribed computational budget may be met with precision. This design renders our token compression process at once principled and adaptive, and thereby attains a robust equilibrium between efficiency and performance.

4.6 COMPUTATIONAL COMPLEXITY ANALYSIS

The computational overhead entailed by our proposed ST-SimDiff framework admits of analysis across three principal stages: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection (SRTS), and Difference-based Event Token Selection (DETS).

The initial stage is concerned with the construction of the graph. For each of the N visual tokens, we compute the cosine similarity with respect to its spatially and temporally adjacent neighbors. Since each token possesses only a small and constant number of neighbors, this stage entails but a single pass over all tokens and consequently incurs a complexity of $O(Nd)$, where d denotes the feature dimension of each token.

Following graph construction, the SRTS module proceeds to identify token clusters. As detailed in our implementation (Section 5.2), we adopt a connected components algorithm, which operates with

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
LLaVA-Video	-	-	-	63.3	-	-	58.2	57.3
Token Retain Ratio $r = 30\%$								
FastV	68.2	58.6	51.2	59.3	48.0	59.9	53.5	51.3
PruMerge	70.2	57.3	52.2	59.9	49.6	60.5	54.7	50.9
FasterVLM	71.3	57.6	51.4	60.1	51.0	61.3	55.8	52.6
VisionZip	68.4	55.9	50.4	58.3	46.9	60.3	53.2	53.0
FrameFusion	74.0	59.8	50.0	61.3	49.7	63.3	56.0	53.0
ST-SimDiff (Ours)	74.7	61.9	53.0	63.2	52.5	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$								
FastV	73.9	61.4	51.3	62.2	50.1	62.2	55.7	54.7
PruMerge	72.3	58.7	52.8	61.3	51.4	63.2	56.9	54.6
FasterVLM	74.0	59.2	51.8	61.7	51.3	62.2	56.4	56.2
VisionZip	72.6	57.7	51.6	60.6	51.3	63.2	56.8	54.2
FrameFusion	74.6	61.2	52.0	62.6	51.7	64.2	57.6	55.8
ST-SimDiff (Ours)	76.2	62.4	52.7	63.8	52.3	64.3	57.9	57.3

Table 1: A comparison of performance across long-form video understanding benchmarks conducted on LLaVA-Video-7B with 64 input frames (%). The strongest performance attained among all methods is emphasized in **bold**.

a near-linear time complexity of $O(N + |E|)$; and since the number of edges $|E|$ is at most $3N$, this reduces to $O(N)$. The ensuing filtering step conducted within each community might, in principle, prove computationally intensive: a naive approach that computed all-pairs similarity within each community c_k would carry a complexity of $O(|C| \cdot |c_k|^2 \cdot d)$. To forestall this eventuality, we impose a constraint upon the maximum admissible size of any community, and in the rare event that a community exceeds a predefined threshold—for example, $|c_k| > \sqrt{N}$ —we partition it. This provision ensures that the filtering process remains efficient and does not come to dominate the overall complexity. Although other mainstream graph clustering algorithms, such as the Louvain and Leiden methods, afford more elaborate community definitions, their computational complexity is correspondingly higher, approaching $O(N \log N)$; we therefore elected to employ the simpler and highly efficient connected components algorithm so as to keep processing overhead to a minimum.

The DETS module, in turn, comprises a single pass through all temporal edges in the graph, the purpose of which is to identify significant changes by comparing token similarities against a threshold; this process carries a linear complexity of $O(Nd)$.

In summary, the total complexity of our ST-SimDiff framework is dictated by these linear-time operations and thus culminates in an overall complexity of $O(Nd)$. This proves substantially more efficient than the quadratic complexity of the self-attention mechanism, $O(N^2d)$, on which the Large Language Model relies. Our method, accordingly, delivers significant token reduction while imposing only a negligible computational footprint relative to the model’s principal inference cost.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Baselines With a view to evaluating the effectiveness of our method comprehensively, we adopt LLaVA-Video (Zhang et al., 2024c) and NVILA (Liu et al., 2024c) as our base models and set our

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
NVILA-Video	-	-	-	61.5	-	-	56.3	52.9
Token Retain Ratio $r = 30\%$								
FastV	69.4	55.3	48.8	57.9	50.8	55.6	53.0	49.7
PruMerge	71.0	53.7	49.9	58.2	49.3	58.1	53.4	47.5
FasterVLM	72.6	56.9	51.0	60.1	48.7	57.8	53.0	49.3
VisionZip	71.4	56.2	49.7	59.1	46.3	56.2	50.9	48.9
FrameFusion	72.1	55.7	48.7	58.8	51.3	59.3	54.9	51.3
ST-SimDiff (Ours)	73.3	57.7	49.7	60.2	51.7	59.2	55.2	51.7
Token Retain Ratio $r = 50\%$								
FastV	71.9	58.3	49.3	58.9	49.5	58.8	53.9	50.2
PruMerge	71.2	54.8	46.8	57.6	49.7	58.7	53.9	48.9
FasterVLM	74.6	57.8	50.1	60.8	48.5	58.1	53.0	50.5
VisionZip	72.8	58.1	50.6	60.5	50.4	58.9	54.4	50.3
FrameFusion	72.7	57.2	48.3	59.4	51.0	59.2	54.8	52.6
ST-SimDiff (Ours)	73.9	59.7	51.4	61.7	52.3	61.3	56.5	52.5

Table 2: A comparison of performance across long-form video understanding benchmarks conducted on NVILA-Video-8B with 64 input frames (%). The strongest performance attained among all methods is emphasized in **bold**.

approach against a spectrum of state-of-the-art video token compression techniques. These baselines span a diverse array of strategies, encompassing **importance-based reduction methods** such as FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b), together with **hybrid reduction methods** such as VisionZip (Yang et al., 2025), FrameFusion (Fu et al., 2024b), and PruMerge (Shang et al., 2024). In order to render the comparison both fair and complete, we further include the original uncompressed model (Vanilla) as a performance upper bound and Random Sampling of an equivalent number of tokens as a corresponding lower bound.

Benchmarks To probe the capabilities of our method in long-video understanding, we enlist three challenging benchmark datasets: VideoMME (Fu et al., 2024a), LongVideoBench (Wu et al., 2025), and EgoSchema (Mangalam et al., 2024). VideoMME constitutes a comprehensive benchmark for foundational video understanding, comprising diverse video types together with three categories of duration, namely Short, Medium, and Long. LongVideoBench appraises the capacity to retrieve and reason about details within long videos by way of a “referring reasoning” task, drawing upon videos of four distinct lengths: 15s, 60s, 600s, and 3600s. EgoSchema, finally, is a question-answering dataset centered upon long-term, first-person videos, and it interrogates the model’s ability to apprehend the intentions of characters and the causal chains that link their actions.

5.2 IMPLEMENTATION DETAILS

We conduct our experiments on NVIDIA L20 GPUs endowed with 48GB of VRAM under Ubuntu 22.04, and the inference evaluation is carried out by means of the *lmms-eval* (Zhang et al., 2024a) library. For LLaVA-Video, we abide by its original configuration of 64 input frames; and in order to secure a fair comparison of computational efficiency and resource consumption under commensurate conditions, we establish a consistent setup by likewise fixing the input frame count for NVILA at 64. With respect to the hyperparameters, we set the similarity threshold to $\tau_{\text{sim}} = 0.8$ and the difference threshold to $\tau_{\text{diff}} = 0.2$. The experiments are conducted at token compression rates of $r = 30\%$ and $r = 50\%$ respectively. In our practical implementation, and in consideration of the trade-off

between time expenditure and performance gain, we elect to employ connected component finding as the community detection algorithm.

5.3 COMPARISON WITH STATE-OF-THE-ARTS

In order to evaluate our proposed ST-SimDiff framework comprehensively, we set it against a range of mainstream efficient video language model (VLM) compression methods. The entirety of these experiments is conducted upon three widely adopted long-form video understanding benchmarks, namely VideoMME, LongVideoBench, and EgoSchema. So as to substantiate the generalization ability and robustness of our method, we report detailed performance figures on two distinct base models, LLaVA-Video-7B and NVILA-8B, under token retain ratios of 30% and 50%. The corresponding experimental results are set forth in Table 1 and Table 2.

(1) ST-SimDiff consistently outperforms all competing methods across the whole of the test configurations. A finding of particular note is that, at a 50% token retain ratio, the overall performance of our method on both base models not only surpasses that of the other compression algorithms but, on certain benchmarks, even attains or exceeds that of the original model operating with 100% of the tokens. This outcome corroborates the effectiveness and the novelty of ST-SimDiff, and it invites a more searching analysis of the strategies and the limitations that characterize the existing baselines.

(2) Among the importance-based baseline methods (FastV and FasterVLM, for example), FasterVLM typically evinces comparatively stronger performance. Their shared limitation, however, resides in an inefficient treatment of the temporal redundancy so prevalent in videos, for they are inclined to retain tokens that, though important, are nonetheless repetitive, and thereby curtail the efficiency of information compression. The hybrid baseline methods (PruMerge, VisionZip, and FrameFusion, for instance) exhibit more competitive performance than do the importance-only approaches, with FrameFusion generally yielding the most outstanding overall results. Even so, the common optimization objective of these methods remains that of identifying and preserving representative “commonality” information, an objective that harbors a theoretical blind spot inasmuch as it may overlook the key narrative information propelled by “turning points” and “abrupt events”.

(3) In sum, while the existing baselines reflect an ever-deepening understanding of “redundancy” in videos, they continue to confront two core challenges: first, an absence of any synergistic analysis of the jointly spatio-temporal similarity between visual tokens, which renders complex redundancies difficult to capture; and second, a general neglect of “difference” detection, which leaves them unable to actively preserve the key changes that define the development of the plot. The performance gain realized by ST-SimDiff arises precisely from its redress of these deficiencies, achieved by constructing a spatio-temporal graph that models complex spatio-temporal relationships within a single unified representation and by innovatively introducing difference detection.

5.4 ABLATION STUDY

In order to substantiate the effectiveness of each core component within our ST-SimDiff framework, we undertook a series of detailed ablation studies. We commenced with a Baseline strategy that incorporates only rudimentary importance-based pruning, and thereupon introduced, in progressive fashion, our proposed similarity selection module (+Sim) and difference selection module (++Diff), the better to quantify their respective contributions. The whole of these experiments was conducted at token retain ratios of 30% and 50%, with the results presented in Table 5.

(1) We first appraised the effectiveness of the +Sim module. The experimental results indicate unambiguously that the introduction of similarity modeling atop the baseline improves model performance in a consistent manner. We further decomposed the similarity strategy into three variants: spatial-only (Spatial), temporal-only (Temporal), and spatio-temporal joint (Spa. + Tem.). The data attest that the collaborative modeling of the spatio-temporal associations between visual tokens is the most effective means of compressing redundancy.

(2) Having integrated the optimal spatio-temporal joint similarity module, we then introduced the difference selection module (++Diff) so as to constitute our complete framework. The addition of this module furnishes a pronounced leap in performance, most conspicuously under the aggressive compression ratio of 30%. This attests to its vital complementary function: whereas the aggressive +Sim module concentrates upon redundancy, the ++Diff module operates as a safeguard by which

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	60.3	56.2	54.8
+ Sim	Spatial	61.5	56.5
	Temporal	61.7	56.8
	Spa. + Tem.	62.6	57.0
++ Diff (Proposed)	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$			
Baseline	63.2	56.7	56.5
+ Sim	Spatial	63.3	57.0
	Temporal	63.7	57.2
	Spa. + Tem.	63.7	57.8
++ Diff (Proposed)	63.8	57.9	57.3

Table 3: Ablation results delineating the contribution of the several components (%).

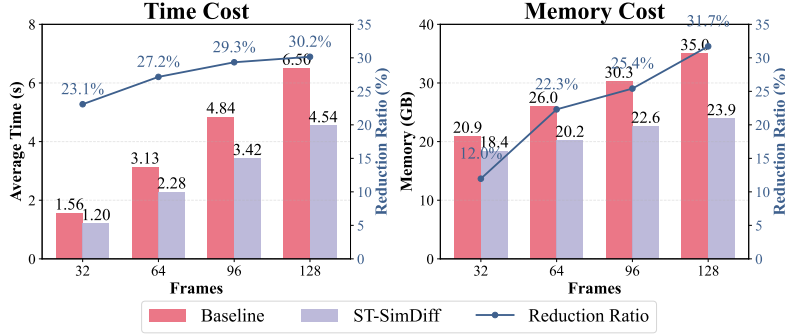


Figure 3: A comparison of computational cost between our method and the baseline LLaVA-Video model, expressed in terms of average inference time and peak GPU memory usage under a 30% token budget.

the unique event tokens that define the video’s narrative are identified and preserved. By contrast, the marginal gain conferred by the module is more modest at a 50% retention ratio, and this is because the more lenient +Sim module is the more likely to have already captured such event tokens, the overall model performance being, moreover, already in the vicinity of its upper bound. This analysis confirms our core motivation: the strength of the framework resides in its capacity to balance similarity-based redundancy compression against the crucial preservation of difference-based events, and it does so most tellingly in demanding high-compression scenarios.

5.5 COMPUTATIONAL COST ANALYSIS

In order to quantify the efficiency advantages that our proposed ST-SimDiff method confers in practical applications, we conducted detailed measurements of its inference time and peak GPU memory usage under a 30% token budget, comparing it throughout against the baseline model (Baseline LLaVA-Video) operating without any compression. The experimental results reported in Figure 3 span video inputs ranging in length from 32 to 128 frames.

Inference Time ST-SimDiff exhibits a substantial acceleration. As the number of input video frames increases, the average inference time of the baseline model climbs steeply, from 1.56 seconds at 32 frames to 6.50 seconds at 128 frames. Our method, by contrast, effectively arrests this growth, requiring an average of merely 4.54 seconds when processing 128 frames. It follows that ST-SimDiff

realizes an ever greater rate of time savings for longer videos, the figure improving from 23.0% at 32 frames to 30.2% at 128 frames.

Memory Cost ST-SimDiff likewise acquits itself exceptionally well. The peak GPU memory usage of the baseline model rises linearly with video length, from 20.9 GB at 32 frames to 35.0 GB at 128 frames. Our method appreciably alleviates this memory pressure by substantially diminishing the number of tokens conveyed to the large language model: when processing 128-frame videos, the peak memory usage of ST-SimDiff amounts to only 23.9 GB, a saving of 31.7% of the memory footprint relative to the baseline. This advantage empowers our model to process longer videos on hardware of limited memory, and thereby markedly broadens the compass of its potential applications.

In sum, ST-SimDiff not only attains leading performance but also manifests significant advantages in computational efficiency, with respect to time and memory alike, thereby establishing its efficiency for the task of video understanding.

6 CONCLUSION

In this paper, we have addressed the computational inefficiency that besets Multimodal Large Language Models in the processing of long videos, a problem whose roots lie at once in the excessive number of visual tokens and in the neglect of critical content changes. We propose ST-SimDiff, a framework that dispenses with any need for training and that constructs a spatiotemporal graph so as to model the complex associations obtaining among tokens. The method avails itself of a dual selection strategy, wherein similarity-based selection identifies redundancy by way of community detection while temporal-difference-based selection preserves the tokens situated at key transition points. Extensive experiments on VideoMME and LongVideoBench demonstrate that ST-SimDiff appreciably reduces both processing time and memory usage while consistently surpassing state-of-the-art performance. This work thereby opens a new research direction, one that balances content representativeness against transitional events in order to arrive at a more efficient representation of video data.

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A VISUALIZATION OF THE TOKEN SELECTION PROCESS

In order to afford a more intuitive appreciation of the ST-SimDiff framework, we present in Figure 4 a selection of visualization samples depicting both the intermediate and the final token selection results. The figure illustrates the process upon a sample video sequence that features a dynamic object manipulation task. Regarding the Similarity-based Representative Token Selection (SRTS) first of all, the rows labeled “Cluster 1” and “Cluster 2” demonstrate the workings of our graph community detection algorithm, which successfully aggregates spatially and temporally redundant regions—the static background and stationary objects among them—into cohesive communities. This visualization confirms that SRTS effectively identifies stable visual elements and compresses them by retaining only a handful of central “representative tokens,” which appear as sparse yellow boxes in the final results; such a mechanism enables the model to accommodate massive background redundancy with considerable efficiency.

Complementing this, the Difference-based Event Token Selection (DETS) discharges a critical role during dynamic moments. As may be discerned in the “Final Results” row, the tokens triggered by high temporal difference are explicitly marked with dense red bounding boxes. These red boxes coincide precisely with the moving hand and the shifting blue liquid, thereby verifying that our method accurately localizes the “turning points” at which the temporal difference exceeds the threshold (τ_{diff}). This guarantees that fine-grained action details and rapid visual changes are preserved, and so permits the model to apprehend the essence of the event without being inundated by static data.

Finally, the synergistic effect of these two components is made evident in the overlay of the final results. The visualization discloses a compelling equilibrium, in which the sparse yellow tokens stand for the stable background while the dense red tokens capture the rapid motion; this contrast furnishes direct visual corroboration of our core motivation. ST-SimDiff circumvents the blind spots of similarity-only approaches by sampling densely during rapid changes, and at the same time prunes those spatiotemporal redundancies that importance-based methods so often fail to detect. Our framework, in consequence, attains an optimal balance between computational efficiency and the preservation of critical visual information.

B COMPARISON RESULTS ON QWEN2.5-VL

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
Qwen2.5-VL-64	74.2	62.7	51.8	62.9	54.2	64.8	59.2	63.0
Qwen2.5-VL-128	77.4	65.8	56.1	66.4	56.2	65.1	60.4	63.9
Token Retain Ratio $r = 30\%$								
ST-SimDiff-64	73.8	61.7	51.8	62.4	52.2	62.1	56.8	62.7
ST-SimDiff-128	75.4	64.7	55.2	65.1	53.1	65.0	58.6	64.4
Token Retain Ratio $r = 50\%$								
ST-SimDiff-64	74.2	62.1	52.2	62.9	54.2	63.2	58.4	62.7
ST-SimDiff-128	76.8	66.3	55.8	66.3	54.9	65.3	59.8	64.5

Table 4: A comparison of performance across long-form video understanding benchmarks conducted on Qwen2.5-VL-7B with 64 and 128 input frames (%).

In order to further substantiate the generalization capability of our proposed framework, we conducted supplementary experiments upon the recently released Qwen2.5-VL model. These evaluations were carried out on the VideoMME and LongVideoBench benchmarks so as to confirm that the efficacy of our method is not confined to a particular model architecture but admits of broad application.

The empirical results, presented in Table 4, attest to the strong performance of ST-SimDiff on the Qwen2.5-VL model. At a token retention ratio of 30%, our method attained scores of 62.4 on VideoMME and 56.8 on LongVideoBench; and when the token budget was enlarged to a 50% reten-

tion ratio, its performance rose to 62.9 on VideoMME and 58.4 on LongVideoBench. It is especially noteworthy that, at the 50% ratio, the performance on VideoMME matches the upper-bound figure achieved with the full complement of tokens, while the score on LongVideoBench is highly comparable to its full-performance counterpart. These findings affirm the excellent generalization ability of our framework.

C ABLATION STUDY RESULTS ON NVILA

Method		VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$				
Baseline		59.3	53.8	50.2
+ Sim	Spatial	59.5	54.5	50.9
	Temporal	59.6	54.3	51.2
	Spa. + Tem.	59.9	54.6	51.3
++ Diff (Proposed)		60.2	55.2	51.7
Token Retain Ratio $r = 50\%$				
Baseline		60.9	55.3	51.5
+ Sim	Spatial	61.1	55.7	51.9
	Temporal	61.3	55.6	51.8
	Spa. + Tem.	61.4	56.0	52.3
++ Diff (Proposed)		61.7	56.5	52.5

Table 5: Ablation results delineating the contribution of the several components on the NVILA-Video-8B model (%).

In order to further substantiate the generalization ability of each component of our framework across disparate base models, we additionally conducted a series of detailed ablation studies on the NVILA-Video-8B model (Liu et al., 2024c). The experimental setup accords with that of the ablation study in the main paper: we commence with a Baseline strategy that incorporates only rudimentary importance-based pruning, and thereafter introduce, in progressive fashion, the similarity selection module (+ Sim) and the difference selection module (++ Diff).

As shown in Table 5, the experimental results once again corroborate the effectiveness of our framework design, and the observed trends prove highly consistent with the performance recorded on LLaVA-Video-7B. First, upon the introduction of the similarity module (+ Sim) atop the baseline, the model’s performance exhibits a steady improvement at both the 30% and the 50% token retain ratios. Within the similarity module, we compared three strategies—spatial-only (Spatial), temporal-only (Temporal), and spatio-temporal joint (Spa. + Tem.)—and the results establish that spatio-temporal joint modeling (Spa. + Tem.) is the most effective of the three. At $r = 50\%$, for instance, the spatio-temporal joint strategy attained a score of 61.4% on VideoMME, thereby outperforming both the spatial-only and the temporal-only strategies. This again demonstrates how important it is to consider spatio-temporal associations collaboratively if redundant information is to be identified effectively. Having integrated the optimal spatio-temporal joint similarity module, we then introduced the dissimilarity selection module (++ Diff) so as to constitute our complete ST-SimDiff framework. The table shows unmistakably that the addition of this module conferred the most significant performance gains. This outcome forcefully demonstrates the effectiveness and the generalization ability of the two core modules of our framework—similarity and dissimilarity selection—and once more vindicates our core idea of addressing redundancy and change simultaneously.

D ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON LLaVA-VIDEO

Within our framework, the similarity threshold τ_{sim} and the difference threshold τ_{diff} constitute two key hyperparameters that govern, respectively, the selection of representative and of transitional

tokens. In order to investigate their impact, we conducted a series of ablation studies, the results of which are shown in Figure 6. The experiments reveal that the influence of both parameters upon model performance follows a comparable trend—rising initially and thereafter declining—while nonetheless exhibiting sound robustness across a certain range. In the case of τ_{sim} , a value set too low occasions imprecise community detection, whereas a value set too high can vitiate the integrity of the semantic clusters. In the case of τ_{diff} , a value set too low may introduce noise on account of over-sensitivity, whereas a value set too high may cause the model to overlook key events. According to the experimental results, the model attains optimal performance at $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$; we therefore adopted this optimal configuration throughout all remaining experiments.

E ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON NVILA-VIDEO

In order to ascertain the robustness of the key hyperparameters in our method, we conducted a series of ablation studies on the NVILA-Video-8B model. As shown in Fig. 6, we evaluated the impact exerted by the similarity threshold, τ_{sim} , and by the difference threshold, τ_{diff} , upon the model’s performance across two benchmarks, namely VideoMME and LongVideoBench.

Impact of Similarity Threshold τ_{sim} . We examined a range of values for τ_{sim} within the interval $[0.6, 0.95]$. The experimental results indicate that the model’s performance remains comparatively insensitive to variation in this threshold. On the VideoMME dataset, a peak performance of 60.2% was attained at $\tau_{sim} = 0.8$; likewise, on the LongVideoBench dataset, performance proved optimal at $\tau_{sim} = 0.8$, reaching 55.2%. Although performance diminished slightly when the threshold was set either too high or too low, the overall fluctuation remained minimal, thereby confirming that our method sustains sound stability across a broad range of similarity thresholds.

Impact of Difference Threshold τ_{diff} . We examined τ_{diff} across the interval $[0.1, 0.4]$. As with τ_{sim} , the model’s performance evinced strong robustness to variation in τ_{diff} . On both benchmarks, optimal performance was attained in the vicinity of $\tau_{diff} = 0.2$, with VideoMME at 60.2% and LongVideoBench at 55.2%. This suggests that setting the difference threshold around 0.2 is most effective for capturing the key events of a video, and thereby maximizes model performance even as tokens are compressed.

In sum, these ablation studies confirm the robustness of our proposed method with respect to its key hyperparameters. The results vindicate our decision to adopt $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$ in the main experiments, inasmuch as these values consistently yield optimal performance across disparate benchmarks and disparate base models alike.

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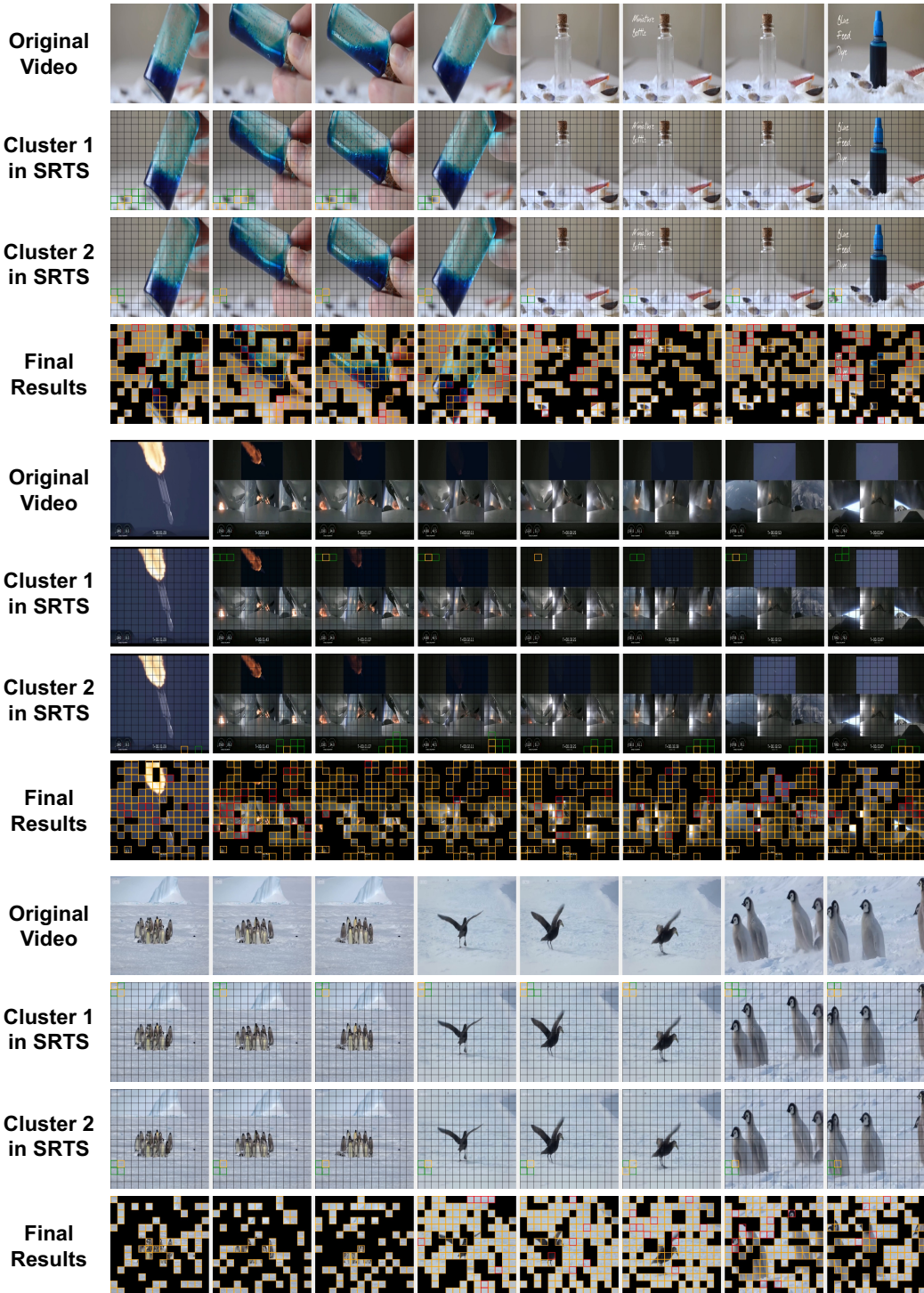


Figure 4: A visualization of the ST-SimDiff process. The visualization decomposes the token selection into the communities detected by SRTS (rendered as grids in Cluster 1 and Cluster 2) and the event-driven tokens detected by DETS. The final result foregrounds the synergy that obtains between the sparse representative tokens (yellow), which encode stable content, and the dense event tokens (red), which encode dynamic actions.

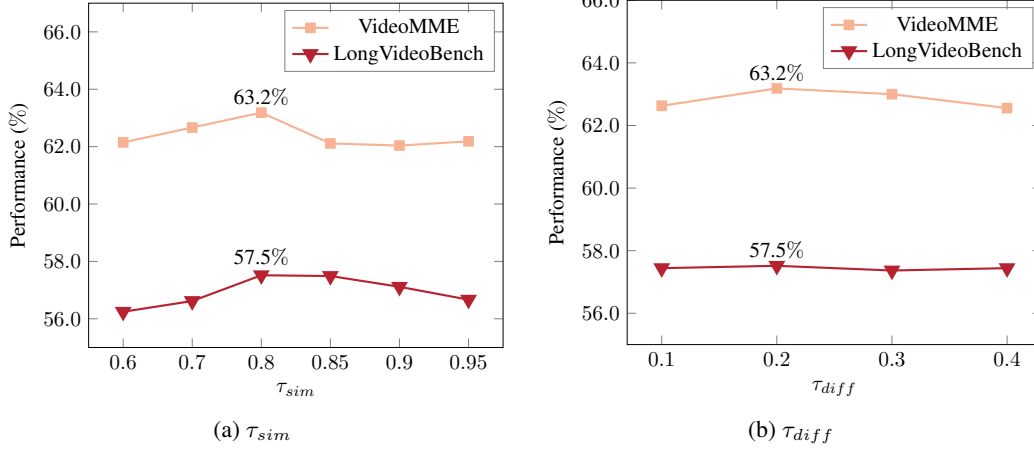


Figure 5: Ablation study results obtained for varying values of τ_{diff} and τ_{sim} on VideoMME and LongVideoBench.

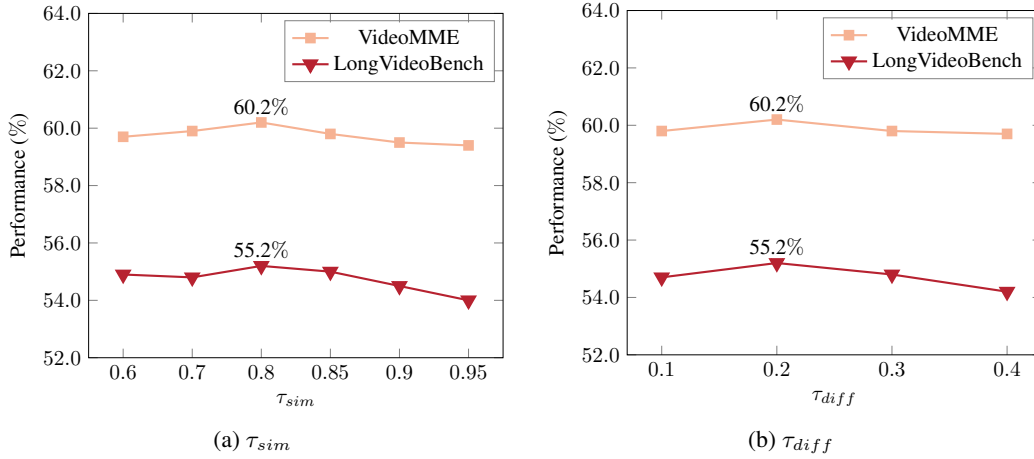


Figure 6: Ablation study results obtained for varying values of τ_{diff} and τ_{sim} on NVILA-Video-8B.